

Reflection Essay – Task 02

This assignment was my first practical experience with single-view metrology, where real-world distances can be estimated from a single photograph using concepts from projective geometry. Before this task, I understood ideas such as homogeneous coordinates, vanishing points, and cross-ratios mostly as mathematical concepts. Working through a complete example helped me understand how these ideas can be combined to solve a real problem.

The first part of the assignment involved estimating the distance between a bicycle and the Arch of Knowledge using only the provided image. I started by measuring the distance between the Main Building reference line and the arch using Google Maps. The measured distance was approximately 160 metres. Unlike the arch, however, the bicycle does not appear as a location that can be directly measured on a map. This immediately highlighted an important limitation: a single image does not provide enough information to recover distances unless additional assumptions or geometric constraints are introduced.

To locate the bicycle, I used the Ultralytics implementation of YOLOv8, a pretrained object detection model trained on the COCO dataset. My first attempt used the YOLOv8s model. The model detected several objects in the image, including a nearby motorcycle, but it failed to detect the bicycle because it was much farther away and occupied only a small number of pixels in the image. After experimenting with different settings and models, I was able to obtain a bicycle detection with a reasonable confidence score. This was an interesting reminder that deep learning models are not perfect and that object detection results often need to be verified rather than accepted blindly.

Once the bicycle had been detected, I drew a blue reference line passing through the points required for the cross-ratio calculation. I used the Preview application on macOS to annotate the image and later used Photopea to obtain pixel coordinates for the selected points. The points represented the camera reference position, the bicycle, the arch, and the vanishing point. Using these coordinates, I calculated the image-space cross-ratio and then used the projective geometry relationship involving the point at infinity to estimate the real-world distance between the bicycle and the arch.

For the second part of the assignment, I repeated the same procedure using my own photograph. Instead of a bicycle, I used a person as the object of interest and selected a shack in the background as the reference object. The distance between the camera position and the shack was measured as 72.25 metres. After detecting the teammate with YOLO, I marked the required points on the image, extracted their coordinates using Photopea, and computed the cross-ratio. Using the same geometric reasoning as before, I estimated the distance between the teammate and the shack to be approximately 28.3 metres.

One of the most valuable lessons from this assignment was understanding how sensitive the final answer can be to point selection. Small differences in the placement of the vanishing point or reference points can noticeably affect the final distance estimate. I also learned that practical computer vision problems rarely work perfectly on the first attempt. The initial failure

of the object detector and the need to manually verify detections reinforced the importance of combining automated methods with human judgment.

Another takeaway was the role of assumptions in projective geometry. The method depends on the selected points being approximately collinear and lying on the same ground plane. In real photographs these assumptions are never perfectly satisfied, which means the final answer is always an estimate rather than an exact measurement. Despite these limitations, I found it remarkable that a single photograph contains enough geometric information to recover meaningful real-world distances.

For the second part of the assignment, I captured multiple photographs before selecting the final image for analysis. I found that small changes in camera position significantly affected the visibility of the vanishing point and the alignment of the points required for the cross-ratio calculation. Some photographs were discarded because the road edges were not clearly visible, while others were rejected because the person was either too close to the camera or too far away for reliable detection. This process helped me appreciate how data collection itself can influence the success of a computer vision task.

The photographs were taken during daylight conditions. I noticed that good lighting made both object detection and manual point selection much easier. I expect the method would become less reliable during dawn, dusk, or night-time conditions because object detectors generally perform worse when objects are poorly illuminated. Similarly, identifying accurate vanishing points and reference points would become more difficult as image contrast decreases.

The second part of the assignment was particularly valuable because it required me to recreate the entire pipeline independently rather than follow a predefined example. Unlike the bicycle image provided in the assignment, my own image required me to decide where to place the reference line, which object to use as a reference landmark, how to measure the known distance, and how to verify the object detector's output. This made the exercise feel much closer to a real-world computer vision problem where the inputs are not perfectly prepared beforehand.

The third part of the assignment was a third-person view, taken from behind the photographer so that the camera, the photographer, the teammate, and the vanishing point lie on the same line. I used the same YOLOv8 model to detect both people and took the bottom-centre of the teammate's bounding box as her ground point. The vanishing point was found by fitting only the avenue's own edges (the hedges and lamp-post rows), since the nearby building edges otherwise biased the estimate. Using the back-gate to front-gate distance of 220 metres from Google Maps as the known reference, the cross-ratio gave a distance of about 192.6 metres between the teammate and the front gate.

Another observation was that object detection and projective geometry complement each other well. YOLO alone can identify what an object is and where it appears in an image, but it does not directly provide real-world distances. Projective geometry, on the other hand, can estimate distances but requires accurately identified image points. Combining the two approaches allowed me to move from simple object recognition to quantitative measurement from a single image, which I found to be one of the most interesting aspects of the assignment.

